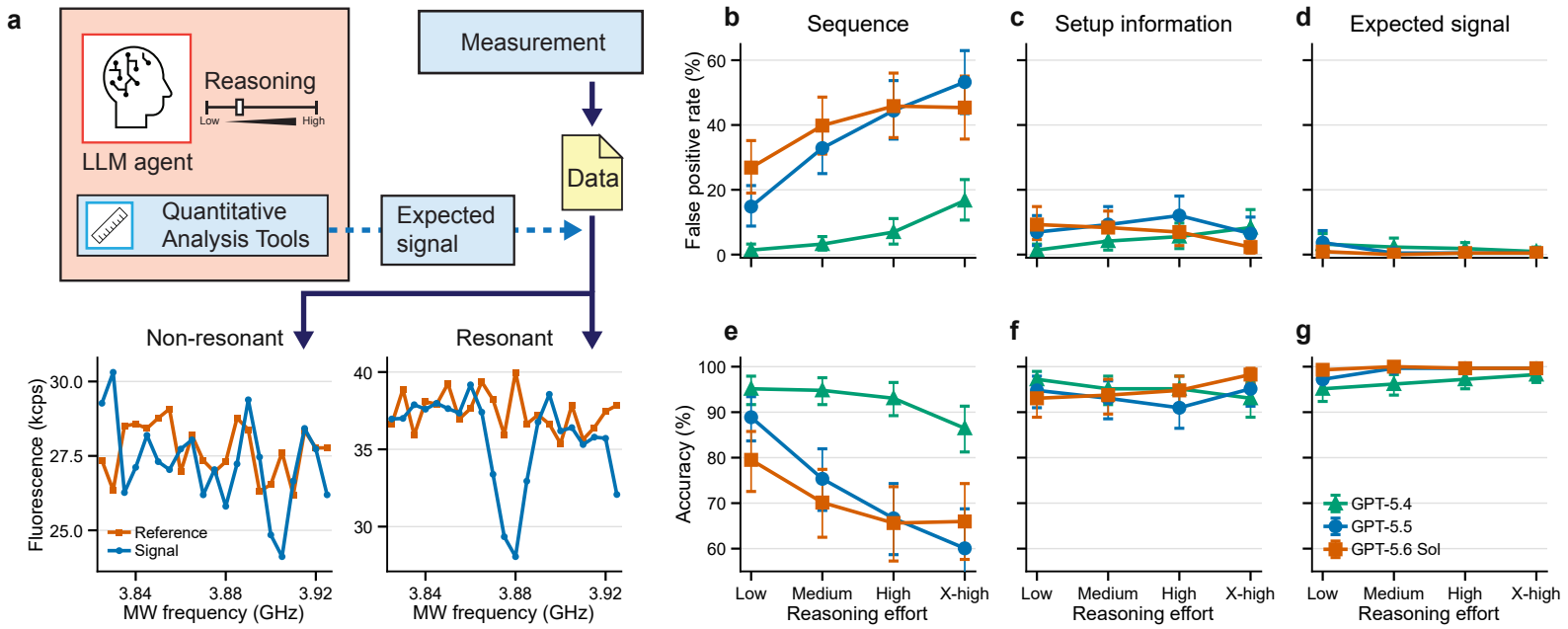

a
Reasoning
Low
High
LLM agent
Quantitative Analysis Tools
Measurement
Data
Expected signal
Non-resonant
Resonant
Fluorescence (kcps)
30.0
27.5
25.0
Reference
Signal
3.84
3.88
3.92
MW frequency (GHz)
40
35
30
b
Sequence
c
Setup information
d
Expected signal
False positive rate (%)
60
40
20
0
e
f
g
Accuracy (%)
100
90
80
70
60
Low
Medium
High
X-high
Reasoning effort
GPT-5.4
GPT-5.5
GPT-5.6 Sol